ISUZU Isuzu Motors America, LLC **Request for Quotations**	Ref # : RFQ - AS - 040			(1 / 4)
	Date : Oct. 14, 2010			Revision 0
	VP	Dir	Mgr	SS
				Doug Walby 10/15/10

1. Title	Adding LSD test process to DVT drive curve
2. Units	1 system
3. Quote due date	October 25, 2010
5. Delivery date	November 30, 2010
6. Install Location	Spartan Motors Plant #5 in Charlotte, MI

7. Specifications

7.1. Project Outline

To design and fabricate software/hardware to receive BOM data from Spartan system

7.2. Project Condition

7.2.1. Project Timing

1) Quotation submission : Oct. 25, 2010
2) Purchase Order issue : Nov. 02, 2010
3) Design approval : Nov. 16, 2010
4) Vendor buy-off : Nov. 30, 2010
5) Delivery at site : Nov. 30, 2010

** Following date is only for reference*

6) Start of production : Dec. 06, 2010

7.2.2. Pick-up & Delivery Location

ISUZU MOTORS AMERICA, LLC
c/o Spartan Motors Plant #5
1055 Mikesell Street
Charlotte, Michigan 48813

7.2.3. Project Contact

ISUZU MOTORS AMERICA, LLC
Mr. Doug Walby
Senior Manufacturing Engineer
Tel : 517-541-3809 Ext. 3
Cell : 810-955-1174
email : douglas.walby@isza.com

7.3. Deliverable

7.3.1. List of Deliverable

1) Program and testing --- 1 unit

7.4. General Condition

7.4.1. Facility Condition

1) Electrical power supply : AC480V ± 10%, 3 phase, 60Hz
AC230V ± 10%, 1 phase, 60Hz
AC115V ± 10%, 1 phase, 60Hz

2) Compressed air supply : 390 - 490 KPa

7.4.2. Paint Specification

1) To be complied with ISZ-A standard paint specifications

7.4.3. General Equipment requirement

1) All general maintenance machine requirements must be easily accessible
2) To be complied with OSHA safety specifications

7.4.4. Equipment Components

1) To be readily available (defined as next day UPS red)

7.4.5. Drawings

1) To be submitted by AutoCAD 2000 format or DXF file

7.5. Scope of Work

7.5.1. Program Design

1) To submit the approval system design documents and to be approved by ISZ-A
   * Not allow the fabrication until the design is approved by ISZ-A

7.5.2. Program Test

1) To test using DVT and vehicle at Spartan
2) Trial condition (DVT side) shall be prepared by BEP

7.5.3. Documentation

1) To provide the "As-Built" drawings and manuals as follows
   - Network wiring
   - Nework configuration
2) To provide three(3) sets of paper copies, One(1) set of electrical copy in CD or DVD

7.5.4. Start-up Support

1) To provide the supervisor for production support
   * Support period will be determined between ISZ-A and the vendor

7.6. Detailed Specifications

7.6.1. Timing of LSD Test

1) LSD Test is performed before ABS speed sensor test in driving curve.
2) LSD Test is applied both w/LSD and wo/LSD vehicles.

7.6.2. Protocol of communication DVT and vehicle

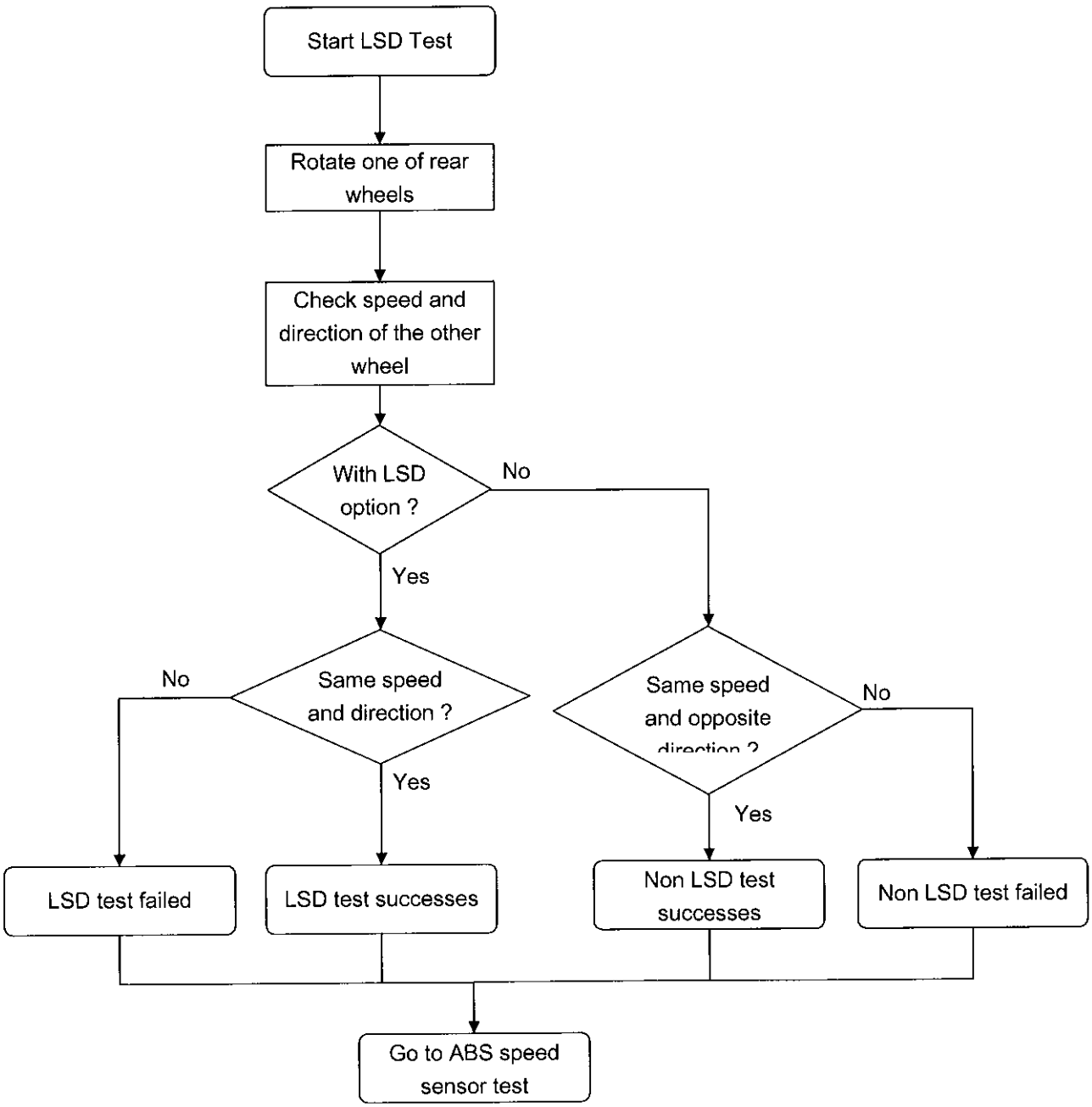
1) ECU communication is not necessary since LSD is mechanical type.

7.6.3. Flow of LSD test

1) DVT rotates one of rear wheels
2) DVT checks speed and direction of the other wheel

with LSD : speed and direction must be same

w/o LSD : speed must be same, direction must be opposite



8. Payment Plan

1) 90% of the payment : After run-off approval at vendor

2) 10% of the payment : After final approval at ISZ-A site

* If this is not an acceptable payment plan, an alternative plan must be clearly stated in the submitted quote.

9. Summary

Items	Responsibility	
	Vendor	ISZ-A
1) Design, Fabrication, Paint	X	
2) Design approval		X
3) Run-off at vendor before shipping	[redacted]	[redacted]
4) Transportation to ISZ-A site	[redacted]	[redacted]
5) Supervising for installation at ISZ-A site	X	
6) Machine run-off at ISZ-A site	X	X
7) Operator, Maintenance training at ISZ-A site	[redacted]	[redacted]
8) Start-up support for start of production	X	
9) Manual and Drawing submission (Approval, As-built)	X	
10) Two(2) years warranty (Mechanical, Electrical)		

End of Specification